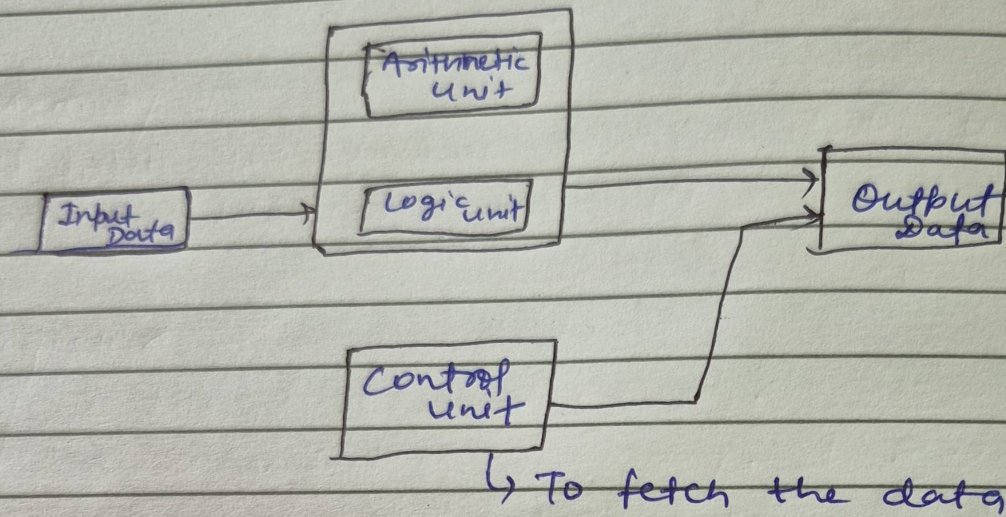


Computer Organisation & Architecture

UNIT:1

Functional Block of Computer



VLSI = very large scale Integrated chips
ULSI = ultra large scale Integrated chips

Memory Read = To fetch data from control unit, for execution. This process is called memory read.

Memory write = After execution of the data then control unit store data in memory and ~~it is~~ this process is called memory write.

Accumulator = It stores the result of Arithmetic Logic unit.

Data = Data can be anything.

For example; Photo, document, audio, video, etc.

pdf = Portable document format
Jpg = Joint photograph group
jpeg = Joint ~~expert~~ photograph ~~expest~~ group

Translator

Compiler

Interpreter

Assembler

Translator is the process to translate one language to another.

Assembler

It is a software which is used for assembly language.

What we write code in Mnemonic assembly language is called Mnemonics code.

ADD R₁ R₂

MOVE A₁ A₂

20/05/26

nibble →

Bit → byte → Kilobyte (KB) → Megabyte (MB) → Gigabyte (GB)
→ Terabyte (TB) → Petabyte (PB) → Exabyte (EB) → Zottabyte (ZB) → Yottabyte (YB)

Smallest unit of memory = Bit

1 nibble = 4 bit

1 byte = 8 bit / 2 nibble

1 MB = 1024 KB / 2¹⁰ KB

1 GB = 1024 MB / 2²⁰ KB

1 TB = 1024 GB / 2³⁰ KB

$$400 \text{ KB/s} = \text{Kilo Byte/second}$$

$$400 \text{ mb} = \text{mega bit/second}$$

$$3 \text{ MB} = 3 \text{ Mega Byte/seconds}$$

Memory

It holds the instruction and data. Without memory the computer will not work properly.

(a) Register

- It is the fastest memory.
- It is a volatile memory, it means when the power is gone the data will be erased.
- The size in Byte.

Types - Accumulator (AC), Program Counter (PC), Instruction Register (IC), Memory Address register (MAR), Status Register (SR) / Flag register

(b) Cache Memory

- It is the very fast memory.
- It is also a volatile memory.
- Size - KB to MB

Types - L1 cache, L2 cache, L3 cache

(c) Random Access Memory (RAM) / Main memory.

- It is also a volatile memory.
- Temporary memory
- Size - 4GB, 8GB, 16GB, 64GB, 32GB.

Types - Static RAM, Dynamic RAM

- SRAM is the fastest memory than DRAM.

Types of DRAM :

- DDR4, DDR5, DDR3
- It refresh the data.

Read Only Memory (ROM)

- It is a non-volatile memory which means the data stored even after the power gone.

Compact Disk (CD)

- It stores 800 MB data.
- It is for only one time use.

SSD = Solid State Drive (256 GB / 512 GB)

HDD = Hard disk drive (500 GB / 1 TB)

- SSD is faster than HDD.

Central Processing Unit (CPU)

- It is brain of computer.
- It is also called a processor.
- It does all processing and decision-making.

Main parts of CPU are -

(i) ALU

It performs arithmetic and logical operation.

100011

011100

1

1001

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Instruction Set Architecture (ISA)

It is a contract between hardware & software and it is the part of CPU that programmer and compiler can see and execute it.

Type —

(a) Instruction Format

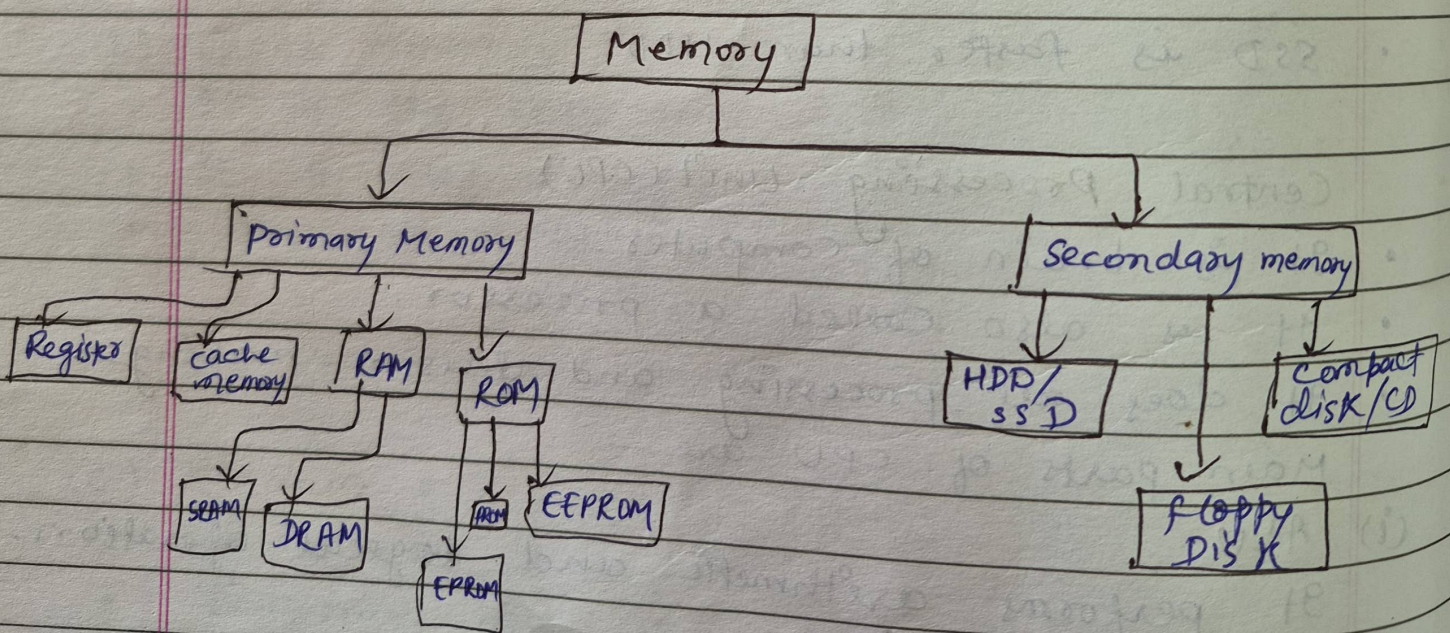
The instruction look in binary.

EX = operation code, Data Register, Source register, immediate register.

(b) The instruction will see what type of operation need to perform like move, load, transfer.

(c) Addressing mode: how the CPU find the operands value.

Immediate, Register, direct, index addressing mode.



Registers

The processor register is the small and high speed storage location built directly into the CPU.

Register holds data, instruction and memory address which is currently process by the CPU.

It is overing register is fast accessing high and memory is hierarchy.

General Purpose Register (GPR)

It can hold other operand or address.

It used for temporary data storage during arithmetic and logical operation.

Data Register

It exclusive hold data value and it cannot be used for calculating memory address.

Address Register

It hold segment or memory address and it is used for specific addressing mode.

e.g - Index Register, Base register and segment pointer

Flag Register

It is used for the hold the previous data which is utilize by the CPU operation. It is also called status register.

gt used for conditional branching.
e.g. carryflag, zero flag, overflow flag.

Program Counter

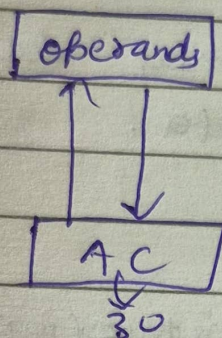
gts contain the address of next instruction to be fetch and execution.

Instruction Register

It keeps temporarily data which specific machine instruction that is currently being decoded or executed.

Accumulator (AC)

It stores the result of ALU operation.



$$5 * 6 = 30$$

Memory Address Register (MAR)

It targets the precise RAM hardware, address slot.

currently access for read or write.

Memory Data Register (MDR)

It operates data which is ^{pulled} ~~poored~~ by the RAM.

Fetch, Decode and Execution Cycle

Fetch, Decode, execution cycle is the fundamental operation that ~~we~~ CPU repeat billions of time per second to run software instruction.

- Fetch

CPU retrieve the raw instruction data from system memory (RAM) and bring into the

Processor.

• Program Counter to MAR

The PC holds the address of the next instruction and the CPU copies this address into the MAR.

- Decode

The CPU interpret that data which control unit bring the data.

• opcode isolation

The CU split that instruction inside the instruction register into two parts - opcode & operand.

The opcode command is - ADD, LOAD, MOV, JUMP.
Operand = The data target on memory address.

- Execution

The CPU carry ~~of~~ ^{or} the command and using its processing component

• ALU operation

It perform arithmetic & logic operation.

• Data transfer

If the instruction is the memory operation like STORE and that data ~~write~~ ^{write} back ^{into RAM} through MAR and MDR.

Instruction Set

An instruction set is the complete collection of all machine level instruction that CPU can execute.

High level: $C = a + b$

Assembly level: Add R_1, R_2

Machine level: 0010 1100 0010

CPU execute it with the help of hardware circuit.

NOTE - Instruction set sits between hardware and software.

Compiler, Assembler and operating system depend on it.

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Addressing Mode

It tell the CPU to specify operand from or how data operands in memory.

Operands can be data, address or register.

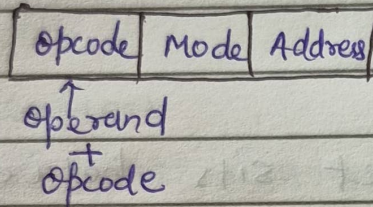
We have 32-bit or 64-bit laptop or device. and we need to initialize the data in memory so we use addressing mode.

Types of Addressing Mode

- | | |
|----------------------|-----------------------------------|
| (a) Implied mode | (f) Immediate mode |
| (b) Indirect mode | (g) Register Indirect mode |
| (c) Direct mode | (h) Auto increment/decrement mode |
| (d) Register mode | (i) Index Register |
| (e) Instruction mode | (j) Base Register |

Implied mode

The opcode definition itself define the operand.

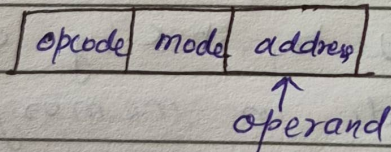


Ex! - Increment Accumulator

Immediate mode

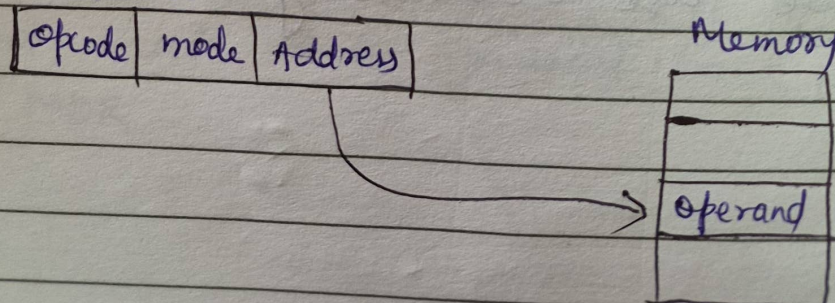
The address field of instruction specify the operand value.

It is used to initialize register with constant value.



Direct mode

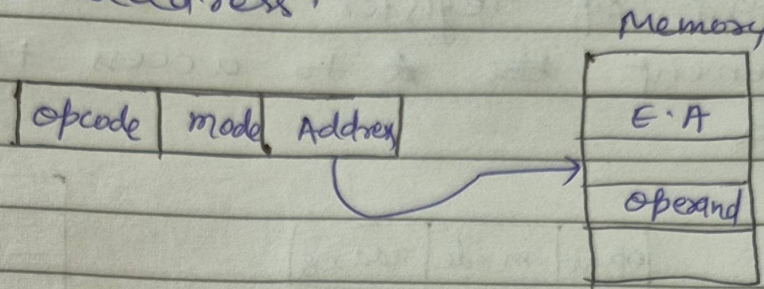
The address field of instruction specify the effective address



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Indirect mode

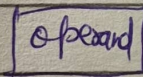
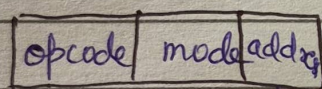
The address field of instruction specify the address of effective address.



- Mode is used to implement pointer.
- The number of memory access to get operand is 2.

Register mode / Register Direct mode

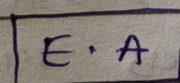
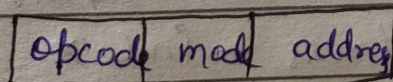
Address field of instruction specify a register which hold operand.



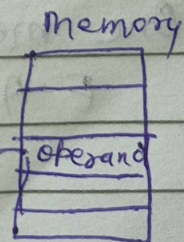
Register [R_1, R_2]

Register Indirect mode

The address field of instruction which specify the register and hold the E.A.

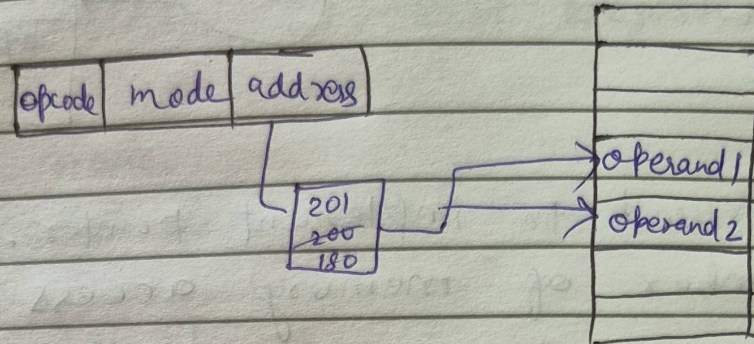


Register (R_1, R_2)



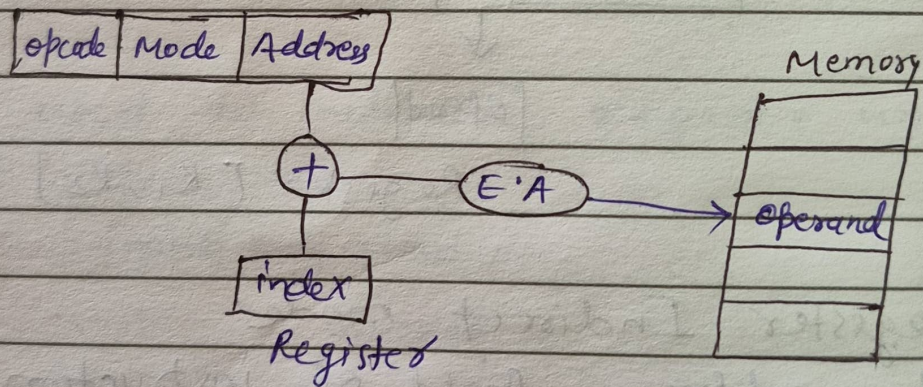
Auto Increment/Decrement mode

Variant of register in direct mode which the content of register is automatically incremented/decremented ~~the~~ to access the sequential data.



Index or Index Register Mode

Base address taken from address field of instruction and index value from index register.



- One memory access get to required operand.
 $E \cdot A = \text{Base address} + \text{index}$

